

STA 6709

Assignment #4

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Executive Summary: Florida Census Tract Area Map

This interactive map displays Florida census tracts using spatial polygon data from the TIGER/Line census tract shapefile. The purpose of the map is to help users explore how census tract land area varies across the state and to identify spatial differences between urban, suburban, coastal, and rural areas.

The map was created using the leaflet package in R and uses the CartoDB.Positron basemap. The polygons are colored by land area group: small tract area, medium tract area, and large tract area.

The map reveals a clear spatial pattern. Smaller census tracts are concentrated in major urban areas such as Miami, Tampa, Orlando, Jacksonville, and other densely populated regions. In contrast, larger census tracts appear more often in rural, less densely populated parts of Florida.

The map also includes pop-up labels for each tract. When users click on a polygon, they can view relevant information including the county FIPS code, tract name, land area in square miles, and area group.

Finally, the map includes layer controls that allow users to turn small, medium, and large tract areas on or off. Overall, the map highlights the contrast between compact urban census tracts and larger rural census tracts, making the spatial structure of Florida easier to understand.